SOUVIND N K

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Data Engineer

PROFILE SUMMARY

- **Experienced Microsoft and Databricks Certified Data Engineer with 5+ years of experience** in building scalable **ETL/ELT** pipelines, data lakehouse solutions, real-time streaming applications and driving project success.
- **Expertise in cloud and data pipeline management & functional development**, testing, development, compilation & data analysis, and diagnostics testing, with a keen focus on ensuring compliance with government regulations and industry standards.
- **Demonstrated leadership in managing cross-functional teams** and leading database system design and development projects from conception to execution, resulting in the successful integration of data engineering and analysis.
- **Strong proficiency in Python programming, SQL programming, data and CI/CD pipeline techniques**, leveraging these skills to streamline processes, reduce manual data compilation time, and improve overall efficiency.
- Hands-on expertise in **Delta Lake advanced features** such as **Time Travel, Liquid Clustering, and Data Security** for governance and compliance. Proficient in **performance tuning, partitioning, and cost optimization** to enhance scalability and efficiency. Strong knowledge of **data governance, lineage, and security best practices** to ensure reliability and compliance.
- Experienced in designing **ETL/ELT pipelines** and implementing **batch processing** as well as **real-time streaming solutions** for large-scale enterprise data. Skilled in **orchestration frameworks** like Airflow and ADF to automate workflows and optimize pipelines and delivering **cloud-native data solutions on Azure and Databricks.**
- **Microsoft Certified Azure data engineer** and **Databricks certified data engineer associate**.

SKILLS

- **Programming Languages:** SQL, Python, PySpark, C, C++, Java
- **Tools:** Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Data Lake, Delta Lake, Unity Catalog, SQL Server, SQL Server Management Studio, Snowflake, Git, Unix, Database, Datawarehouse, Azure Devops, Kafka, Apache Spark, Azure Logic Apps
- Data governance, Data lineage, Data Orchestration, CDC, Z-Ordering, Time Travel, Liquid Clustering, Cost Optimization, Data Security.
- **Soft Skills:** Strong problem solving and analytical skills, good communication, Fast Learner, Empathetic team player, Leadership Time Management, Adaptability, Creativity, Interpersonal skills

WORK EXPERIENCE

Capgemini Technology Services India Limited

Data Engineer | Sept 2024 – Current | client - Mercedes-benz

- Collaborated with data engineers to develop data pipelines and ETL process to improve quality and accessibility, Led the development and implementation of data lake, resulting in 50% increase in data accessibility.
- Spearheaded strategic initiatives to enhance cloud functionality & performance, leveraging innovative technologies & methodologies. Collaborated with stakeholders to define project requirements, manage resources effectively, & drive project success.
- Developed and implemented data pipelines to improve data quality, resulting in 30% increase in data accuracy.
- Evaluated business needs, objectives, interpreted trends and patterns to build data systems and pipelines.
- Implemented and managed CI/CD pipelines to automate development and integrate test pipelines.
- Collaborated with data scientists and architects and identified opportunities for data acquisition and data exploration.
- Developed multiple Pyspark notebooks in databricks and pipelines in azure data factory improving data flow efficiency by 27%.
- Designed and implemented 30+ data pipelines, improving data flow efficiency by 27% and Managed and maintained a database of over 1.5 million records with 99.9% uptime, automated ETL processes, leading to a 47% reduction in manual data handling tasks.
- Enhanced data quality by implementing automated cleansing procedures, reducing errors by 22%, and Migrated 3+ terabytes of data to cloud, resulting in a 33% reduction in infrastructure costs, Improved query response time by 32% through performance tuning.
- Developed and maintained scalable data pipelines and builds out new API integrations to support continuing increases in data volume and complexity and improved data models that feed business intelligence tools.
- Developed multiple Pyspark notebooks in databricks and pipelines in azure data factory improving data flow efficiency.

Tata Consultancy Services

Data Engineer | Feb 2020 – Feb 2024 | client - HP

- Collaborated with analytics and business teams to improve data models that feed business intelligence tools, increasing data accessibility and fostering data-driven decision making across the organization.
- Performed data analysis required to troubleshoot data related issues and assisted in the resolution of data issues.
- Enhanced data quality by implementing automated cleansing procedures, reducing errors by 22%, and Migrated 3+ terabytes of data to cloud, resulting in a 33% reduction in infrastructure costs, Improved query response time by 32% through performance tuning.
- Implemented and managed CI/CD pipelines to automate development and integrate test pipelines.
- Collaborated with data scientists and architects and identified opportunities for data acquisition and data exploration.
- Defined company data assets (data models), spark, sparkSQL jobs to populate data models and designed data integrations.
- Develop, implement and optimize stored procedures and functions using T-SQL, optimizing and maximizing operational efficiency.
- Creation of scripts, complex functions, and triggers to support the quality and integrity of data in databases.
- Creating documentation for database applications and Implementing database authorization and authentication.
- Implemented processes and systems to monitor data quality, ensuring production data is always accurate and available for key stakeholders and business processes that depend on it.
- Worked in a team driving successful Azure implementations and Specialized in Azure services, including Data Factory, Databricks, Synapse, and Snowflake. Focused on cost optimization (30%), and got Client appreciation Awards for the same
- Helped in data quality, and performance improvement of data pipelines. Contributed to the integration of APIs and batch Processing pipelines in Azure.

CERTIFICATIONS & COURSES

- Microsoft Certified Azure Data Engineer Associate (DP 203).
- Databricks Certified Data Engineer Associate.
- Microsoft Certified Azure Fundamentals (AZ 900)
- Fundamentals of Databricks Lakehouse Platform Accreditation.
- Introduction to Generative AI by Google.
- HackerRank Advanced SQL Certification.
- HackerRank Python Basics Certification.
- Smart India Hackathon Participation.
- SQL Bootcamp Certification.

INTERNSHIPS & TRAININGS

- Data + AI Summit 2024 by Databricks | June 12-13**.**
- Powering Next Gen Customer Experience with Gen AI by Databricks.
- Building Data Applications Using Snowflake Marketplace, Snowpark with Python | 03 July 2024.
- Best of Snowflake Summit | 27 June 2024.
- Mobilize Data, Apps and AI Products from Snowflake Marketplace | June 11, 2024.
- Microsoft Research-IISc AI Seminar Series | February 2, 2023.
- Foundations of causal inference and its impacts on machine learning webinar by Microsoft | December 3, 2020.
- Academic Research Summit 2019 on Data Science and AI | January 2019 by Microsoft | Chennai.
- Student Internship | Dinesh Information Technology Systems | June 2017.

PROJECTS

Data Engineering Projects:

- Developed multiple Pyspark notebooks in databricks and pipelines in azure data factory and extracted, cleansed, transformed and loaded data into data lake.
- Partnered with clients on designing the ETL workflows and responsible for designing and implementing back end aggregated tables for Power BI dashboards.

CDC-Based Incremental Data Pipeline for E-Commerce:

- **Technologies:** Databricks, PySpark, Delta Lake, Azure Data Lake Storage (ADLS), SQL, Azure Data Factory, Git/GitHub.
- Built an **incremental ETL pipeline** using **Change Data Capture (CDC)** to process updates from e-commerce transactional data.
- Captured **new, updated, and deleted records** from source tables and maintained **SCD Type 2 dimension tables** for historical analysis. Stored raw and processed data in **Delta Lake**, enabling **time travel queries and versioning**.
- Designed an ETL pipeline in Airflow to clean, deduplicate, and migrate data to SQL Server.
- Automated daily pipeline execution and monitoring, ensuring **accurate and analytics-ready datasets** for reporting and BI dashboards.

Real-Time IoT Sensor Data Pipeline:

- **Technologies:** Kafka, Spark Structured Streaming, Delta Lake, Azure Event Hubs, Python
- Developed a **real-time IoT sensor data pipeline** capturing temperature and humidity data from multiple devices.
- Processed **streaming data with Spark Structured Streaming** and stored in **Delta Lake** for historical analysis.
- Built windowed aggregations and alerts for threshold breaches.
- Enabled **real-time monitoring dashboards** and historical trend analysis for operational insights.

Cloud Data Lakehouse for Stock Market Analysis:

- **Technologies:** Databricks, Delta Lake, PySpark, Azure Data Lake Storage, SQL, Power BI
- Built a **cloud-based data lakehouse** for historical and intraday stock market data.
- Ingested CSV and API data into **Bronze Delta tables**, cleaned and transformed into **Silver tables**, aggregated metrics into **Gold tables** for analysis.
- Enabled **time-travel queries, incremental updates, and partitioned storage** for fast analytics.
- Connected processed datasets to **Power BI dashboards**, providing actionable insights for stock trends and portfolio performance.

College Projects:

- **Third Eye** | Assist the blind person with the access to applications
- **Labour Management System** | Stores the Labour Management Details

EDUCATIONS

- Bachelor of Computer Science Engineering | College of Engineering Thalassery, Kerala, India | 2018 | GPA: 8.12/10

HONORS & AWARDS:

- Best Team of the Quarter Award.
- Client appreciation award from Westpac Banking Corporation.
- Spot Award / Instant Recognition for quick problem-solving or initiative.
- Best Team Player Award for collaboration across teams.
- Innovation Award for introducing new tools or improving workflows
- Shining star award from Capgemini.
- Certificate of excellence.